Aditya Velpula

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Portfolio

F1 STEM OPT Eligible

WORK EXPERIENCE

AI Engineer -- DAPSE Capstone | George Mason University | Backend & Data Feb 2026 – Present
NSI (National Security Innovations Inc.) *Arlington, VA*

- Designed and developed end-to-end web application interfaces for DAPSE, improving usability of policy intelligence workflows for JAG officers
- Built interactive UI components and dashboards for policy search, risk scoring, and legal brief generation
- Created wireframes, prototypes, and UI flows to improve user navigation and decision-making efficiency
- Integrated frontend with backend APIs using RESTful services, enabling real-time query responses and visualization
- Collaborated with cross-functional teams to ensure consistent UX across modules
- Applied component-based design principles and structured UI architecture for scalability
- Contributed to backend system (FastAPI, FAISS, RAG pipelines) while ensuring seamless frontend-backend integration.

Graduate Teaching Assistant Aug 2025 – Present
George Mason University *Fairfax, VA*

- Selected by Professor Harry J. Foxwell to support 2 graduate sections in AIT-580, improving assignment clarity and reducing regrade requests.
- Mentored 50+ students on Python, SQL, R, and AWS Cloud; collaborated with faculty to enhance course materials through data-driven insights.

AI Engineer Nov 2023 – Jul 2024
Indgeos Geospatial *Telangana, India*

- Designed and developed the company website UI, focusing on responsive and user-friendly layouts
- Implemented HTML, CSS, and JavaScript-based frontend components for geospatial data presentation
- Collaborated with backend teams to integrate APIs and dynamic data rendering
- Improved user interaction through clean UI design and structured navigation flows
- Applied version control (Git) and testing practices to maintain UI reliability
- Built scalable frontend structure aligned with modern web standards.

PROJECTS

Obesity Risk Analytics Using AWS Cloud Infrastructure *(Under the Supervision of Prof. Harry Foxwell)*

- Built end-to-end AWS pipeline (S3, RDS/MySQL, Glue) for CDC health datasets; performed SQL-based statistical analysis and Python/R visualizations (Dash, Seaborn, ggplot2).

U.S. Electricity-Rate Analytics: DIKW-Driven Exploration of IOU vs. Non-IOU Pricing

- Analyzed 320K+ records (2020–2023) using Python and SQL; applied regression and clustering to identify geographic and sector-based pricing differences.

Wildfire Risk Prediction Using Climate Data & Machine Learning

- Integrated MODIS/NOAA satellite datasets and trained Random Forest/XGBoost models, achieving AUC of 0.99 for high-accuracy wildfire risk classification.

SKILLS

- **Languages:** Python, C/C++, Java, SQL, R | **Software Engineering:** Algorithms, OOP, Testing, Debugging, Version Control
- **Frontend & UI Development:** HTML5, CSS3, SASS, JavaScript, React.js, Angular, Responsive Design, Component-Based Architecture
- **UX & Design:** Wireframing, Prototyping, Interaction Design, Design Systems, UI/UX Principles, Usability Optimization
- **Cloud & Infrastructure:** AWS (S3, RDS, Glue), Azure (Data Factory, Databricks), FastAPI, SQLite, FAISS, GCP (Basic)
- **Data & ML:** ETL/ELT Pipelines, Data Modeling, Data Warehousing, Pandas, NumPy, PySpark, Scikit-learn, Feature Engineering
- **AI & NLP:** LLMs, RAG Pipelines, Vector Databases, Prompt Engineering, OpenAI API, Langfuse, Multi-Tier LLM Verification
- **Tools:** Git, Jupyter, Linux CLI, Power BI, Docker, Spark (Basics), Airflow (Basics)

EDUCATION

Masters, Data Analytics 2024 – 2026
George Mason University, Fairfax, Virginia

B.Tech, Computer Science and Engineering **(Artificial Intelligence & Machine Learning)** 2020 – 2024
St. Peters Engineering College, Telangana, India

CERTIFICATIONS

- **AWS Certified:** Generative AI Developer (**Professional, Early Adopter**) | ML Engineer (**Associate**) | Data Engineer (**Associate**)
- **AWS Certified:** Solutions Architect (**Associate**) | AI Practitioner (**Foundational**) | AWS Academy: Data Engineering, Cloud Foundations
- AI For Everyone | SQL (Advanced) Certificate